Neural Architecture Search-Based Few-Shot Learning for Hyperspectral Image Classification

Fen Xiao, *Member, IEEE*, Han Xiang, Chunhong Cao, *Member, IEEE*, and Xieping Gao, *Member, IEEE*

***Abstract*— Few-shot learning (FSL) has achieved promising performance in hyperspectral image classification (HSIC) with few labeled samples by designing a proper embedding feature extractor. However, the performance of embedding feature extractors relies on the design of efficient deep convolutional neural network (CNN) architectures, which heavily depends on expertise knowledge. Particularly, FSL requires extracting discriminative features effectively across different domains, which makes the construction even more challenging. In this article, we propose a novel neural architecture search-based FSL model for HSI classification (HCFSL-NAS). Three novel strategies are proposed in this work. First, a neural architecture search (NAS)-based embedding feature extractor is developed to the FSL in HSIC, whose search space includes a group of proposed multiscale convolutions with channel attention. Second, a multisource learning framework is employed to aggregate abundant heterogeneous and homogeneous source data, which enables the powerful generalization of the network to the HSIC with only few labeled samples. Finally, the pointwise-based cross-entropy (CE) loss and the pairwise-based adaptive sparse loss are jointly optimized to maximize interclass distance and minimize the distance within a class simultaneously. Experimental results on four publicly hyperspectral datasets demonstrate that HCFSL-NAS outperforms both the exiting FSL methods and supervised learning methods for HSI classification with only few labeled samples. The code is available at https://github.com/xh-captain/HCFSL-NAS.**

***Index Terms*— Few-shot learning (FSL), hyperspectral image (HSI) classification, multisource learning, neural architecture search (NAS).**

I. Introduction

HYPERSPECTRAL images (HSIs) contain rich spatial and spectral information, which provides discriminative features for ground objects and has been widely used in geological exploration, target detection, and military reconnaissance [1], [2], [3], [4]. HSI classification (HSIC) is a fundamental task aiming to assign a specific category to each pixel [5], [6], [7]. Over the past decades, deep learning (DL) has shown a remarkable ability to extract effective and hierarchical features for different visual tasks. Models such as stacked autoencoders (SAEs) [8], deep belief networks (DBNs) [9], and CNNs [10], [11], [12], [13], [14] have demonstrated satisfactory performance on HSIC. It is well known that the performance of DL-based methods heavily relies on the sufficient number of labeled samples for each class [15], [16], while obtaining labeled samples of hyperspectral image data is expensive and hard.

To deal with the challenge of insufficient labeled samples, various unsupervised [17] and semisupervised [18], [19] learning methods have been explored. Researchers primarily focus on expanding the model's training dataset and improving its performance by relying on the reliability of pseudolabels, which cannot be guaranteed. It is encouraging that few-shot learning (FSL) offers a more efficient approach, where metric-based approaches [20], [21], [22], [23] are widely used as it is an efficient way to classify the target domain with a small amount of labeled data by learning transferable metaknowledge from source domains with a large number of labeled data.

Existing FSL-based methods in HSIC can be roughly categorized into two groups: 1) model-based methods and 2) domain-based methods. Model-based approaches pay attention to the strategy of a handcrafted model by experts' professional knowledge and experience. The strategies include a deep residual 3D-CNN [24], a lightweight cross-scale convolutional network [25], and a classwise attentive embedding network [26]. Domain-based methods attempt to reinforce metaknowledge through diverse strategies of source-domain learning. The transferable feature is learned from another similar HSI set [27] or a natural image set [28]. For HSIC with limited labeled samples, the above methods made great progress by designing a feature embedding module to map all domain data into an embedding space. However, designing a general and efficient embedding model for the diverse domain data is a serious challenge.

Fortunately, a promising solution to address the aforementioned challenge arises from the growing development of neural architecture search (NAS) [29], [30], [31], [32], [33], [34], which aims to automate the architecture design process of CNNs. In the context of HSIC, some studies [35], [36], [37] have been proposed and made great progress by automatically

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Fen Xiao, Han Xiang, and Chunhong Cao are with the MOE Key Laboratory of Intelligent Computing and Information Processing, Xiangtan University, Xiangtan 411105, China (e-mail: xiaof@xtu.edu.cn; 202021002370@smail.xtu.edu.cn; caoch@xtu.edu.cn).

Xieping Gao is with the College of Information Science and Engineering, Hunan Normal University, Changsha 410006, China (e-mail: xpgao@hunnu.edu.cn).

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exploring high-performance network architectures suitable for datasets with ample labeled samples. Their obvious advantage is that they form a more efficient search space through well-designed convolution operators. However, the exploration of NAS in the context of FSL with limited labeled samples still remains challenging.

To address the aforementioned limitations, we propose a novel FSL method for HSIC that can automatically search the optimal architecture for the embedding feature extractor with multisource learning. First, we automatically search the optimal architecture for the embedding feature extractor by designing a new efficient search space that consists of multiscale convolutions with channel attention. Second, a multisource learning framework that incorporates available heterogeneous natural images and homogeneous remote sensing images is proposed to augment the generalization ability of the model. Finally, for further optimization of the feature extractor, we combine the pointwise-based CE loss and the pairwise-based adaptive sparse loss to enhance the interclass distance while minimizing the intraclass distance. In summary, the primary contributions of our work can be summarized as follows.

1) The NAS for FSL in hyperspectral classification is explored for the first time. By designing a search space with attention-based multiscale lightweight convolutional operators, the automatic design of an embedding extractor for HSIC with limited labeled samples is achieved.
2) A multisource learning framework is implemented within the source domain, integrating heterogeneous and homogeneous datasets to enrich the learnable meta-knowledge and enhance its effective transfer.
3) By integrating pointwise-based CE loss and the pairwise-based adaptive sparse loss, enabling the embedding extractor to learn a metric space where sample features exhibit small intraclass spacing and large interclass spacing.

The remainder of this article is organized as follows. Section II presents the related work. Section III describes our approach in detail. Section IV reports and discusses experimental results with state-of-the-art competing methods on four benchmark hyperspectral datasets. Finally, this work is summarized in Section V.

II. Related Works

A. DL Methods for HSIC

DL has been known as an effective feature extraction approach in the past few years. Many DL-based HSIC methods have been proposed by adopting or designing deep classifiers. For example, Chen et al. [8] utilized a single-layer autoencoder and a multilayer SAE to respectively learn shallow and deep features from hyperspectral data. Chen et al. [9] designed a classification strategy based on a DBN. Wei et al. [10] presented a simple CNN architecture containing five layers with weights for HSIC. To enhance classification performance [38], [39]. He et al. [40] proposed a multiscale 3D deep CNN (M3D-DCNN) for HSIC to meet the multiscale feature. Furthermore, to learn abstract-level feature representations, Roy et al. [41] introduced a hybrid multiscale network that combines 3D-CNN with 2D-CNN. Roy et al. [42] introduced an improved deep pyramidal residual network (A^2S^2K-RSNet) to capture discriminative features for HSIC. Xue et al. [43] utilized two subnetworks with shared weights to introduce a Siamese network (S3Net) for HSIC. However, the effectiveness of these methods heavily depends on the availability of sufficient and accurate labeled samples. Meanwhile, an important issue that we need to consider is that overfitting may be likely to occur [44] for limited labeled training samples.

B. FSL Methods for HSIC

In recent years, FSL has gained prominence with limited labeled samples. Many FSL methods have been proposed for hyperspectral image classification to improve performance including feature extractor design, similarity, and relevance metrics. Liu et al. [24] proposed a deep FSL (DFSL) method to train a deep residual 3-D CNN to learn a metric space. However, there is a domain shift between the source and target domains. To address this issue, Li et al. [27] proposed a deep cross-domain FSL method (DCFSL), which trained feature extractor for both the source and target domains and designed a conditional adversarial domain adaptation strategy to overcome domain shift. Li et al. [45] utilized the model-agnostic metalearning algorithm and cutout augmentation to improve HSIC accuracy with limited samples. Moreover, Xi et al. [25] proposed a novel FSL framework for HSIC with a class-covariance metric (CMFSL), which learned global class representations for each training episode by interactively using training samples from the source and target domains. Wang et al. [28] proposed a heterogeneous FSL method (HFSL), which leveraged mini-ImageNet as a heterogeneous source domain and integrated VGG and 1-D spectral convolutional modules to learn the transferable meta-knowledge. Li et al. [46] proposed an FSL classification framework with self-supervised learning to make full use of the abundant labeled data in natural images and the few labeled data in HSI. Currently, Wang et al. [26] proposed a classwise attentive cross-domain FSL framework (CA-CFSL), in which a 3-D residual convolutional embedding model was learned to extract features with discriminability and domain invariance. Bai et al. [47] designed a 3-D local channel attention residual network based on adaptive subspaces and featurewise transformations to avoid model overfitting and mitigate the impact of cross-domain. However, existing FSL-based methods face a critical bottleneck—the manual design of feature extractors. This manual design process not only introduces inefficiencies but also struggles to cope with the diversity of labeled samples in source domains.

C. NAS Methods for HSIC

Within the realm of NAS, there are primarily three prevailing strategies: evolutionary algorithms (EAs) [33], reinforcement learning (RL) [32], and gradient-based architecture search. The EA- and RL-based methods involve training numerous subnetworks during the search phase to identify

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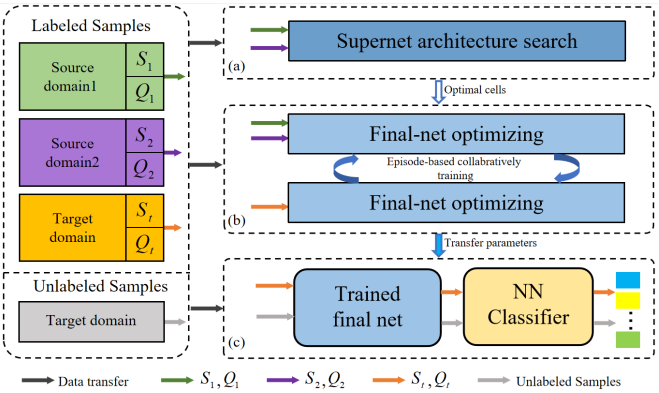


Fig. 1. Workflow of the proposed HCFSL-NAS. (a) NAS for embedding feature extractor. (b) Training in multisource and target domains. (c) Test in the target domain.

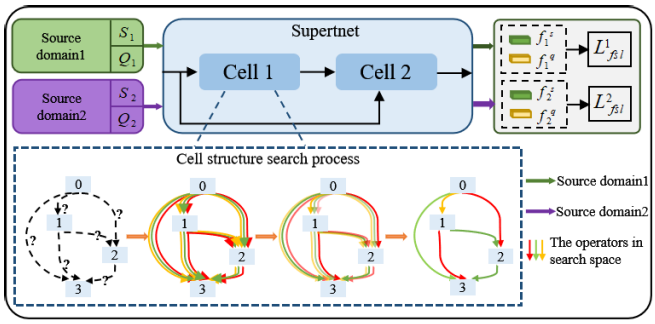


Fig. 2. Architecture search for embedding feature extractor in HCFSL-NAS.

TABLE I
COMPARISON OF PARAMETERS AND FLOPS FOR DIFFERENT CONVOLUTIONS

Operator	Convolution	Parameters	FLOPs
$conv(3\times3)$	$conv(3\times3)$	$9\times C^2$	$9\times C^2\times H\times W$
$conv(5\times5)$	$conv(5\times5)$	$25\times C^2$	$25\times C^2\times H\times W$
$conv(7\times7)$	$conv(7\times7)$	$49\times C^2$	$49\times C^2\times H\times W$
$Spatial_sepconv_(3\times3)$	$conv(1\times3)(conv(3\times1))$	$6\times C$	$6\times C\times H\times W$
$Spatial_sepconv_(5\times5)$	$conv(1\times5)(conv(5\times1))$	$10\times C$	$10\times C\times H\times W$
$Spatial_sepconv_(7\times7)$	$conv(1\times7)(conv(7\times1))$	$14\times C$	$14\times C\times H\times W$

the optimal architecture. To optimize the search process, Liu et al. [34] proposed DARTS, a gradient-based method that trains a supernet comprising multiple cells during the search phase. The optimal subnetwork is selected from this supernet, reducing the computational intensity and making the process more efficient compared to EA- and RL-based methods.

In the field of HSIC, DARTS [34] can automatically design high-performance network architectures according to datasets when a sufficient number of labeled samples are available for each class [35], [36], [37]. For instance, Chen et al. [35] constructed a search space based on 1-D and 2-D convolutional operators and searched for the optimal classification network, achieving impressive classification performance. Zhang et al. [36] further explored the structural characteristics of HSI and designed a 3-D asymmetric search space and proposed a classification model 3D-ANAS, significantly enhancing the classification performance. In addition, to reduce model parameters and facilitate spectral–spatial feature fusion, Cao et al. [37] designed a new lightweight efficient search space and constructed a lightweight classification model LMSS-NAS, which achieved improved classification performance and low computational costs. To enhance HSI classification performance and maximize the utilization of available samples, Wei et al. [48] introduced a two-head semisupervised HSI classification framework based on DARTS, incorporating both labeled and unlabeled data in the training process. The DARTS-based methods efficiently search for network architectures suitable for specific tasks with ample labeled samples. While being effective with ample labeled samples per class, DARTS-based methods need further exploration when dealing with limited labeled samples in real-world scenarios.

III. PROPOSED METHOD

In this study, we propose an automatically searched FSL model called HCFSL-NAS for HSIC with few labeled samples. The workflow of HCFSL-NAS is presented in Fig. 1, which consists of three main components: NAS for embedding feature extractor, training in multisource and target domains, and testing in the target domain. In the FSL process, the labeled samples in the source and target are divided into support set S and query set Q, respectively. First, as shown in Fig. 1(a), we construct a supernet based on NAS algorithms with multisource-domain datasets, automatically searching for the optimal architecture of embedding feature extractor. Subsequently, as illustrated in Fig. 1(b), we utilize the best-designed embedding feature extractor (final-net) obtained from part (a) for alternating optimization training, which involves a substantial number of labeled samples from the multisource domain along with a limited number of labeled samples from the target domain. Finally, as depicted in Fig. 1(c), we use a nearest neighbor (NN) classifier to categorize the features extracted from the target-domain test samples by the optimized final-net.

A. NAS for Embedding Feature Extractor

1) Search Space: To search for the optimal embedding feature extractor, the search space needs to be determined first. Inspired by LMSS-NAS [37], we design a multiscale lightweight search space combining spatial separable convolution [49] for the diversity of feature scale in hyperspectral image, and Table I presents the parameters and floating-point operations (FLOPs) of the spatial separable convolution and conventional convolution with the same receptive field. In spatial separable convolution (i.e., $Spatial_sepconv_(k\times k) = \text{conv}(1\times k)((\text{conv}(k\times 1)(x)))$, the key idea is to replace conventional filters with separable filters to reduce the parameters while maintaining same receptive fields. For instance, suppose that the input feature x has a size of $C_{\text{in}}\times H\times W$ and the output needs to be $C_{\text{out}}\times H\times W$; the number of parameters and FLOPs in conventional convolution (i.e., $\text{Conv}(k\times k) = \text{conv}(k\times k)(x)$) are computed by $P_{cc} = C_{\text{in}}\times k\times k\times C_{\text{out}}$ and $F_{cc} = C_{\text{in}}\times k\times k\times C_{\text{out}}\times H\times W$. Taking $C_{\text{in}} = C_{\text{out}} = C$, the parameters P_{sc} and FLOPs F_{sc} in spatial separable convolution are just $2/(k\times C)$ of conventional convolution, where $P_{sc} = C_{\text{in}}\times k\times 1 + C_{\text{out}}\times 1\times k$ and $F_{sc} = C_{\text{in}}\times k\times 1\times H\times W + C_{\text{out}}\times 1\times k\times H\times W$. At the

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same time, in order to enhance the adaptive feature extraction ability [50] of the convolution operator, we adopt the channel attention map M_c to strengthen the adaptive learning ability on the spectral features. The search space $O = \{o^1, o^2, \ldots, o^7\}$ is designed as follows:

- None : $f(x) = 0$
- avg_pool_3 × 3 : AvgPool2d
- max _pool_3 × 3 : MaxPool2d
- skip_connect : $f(x) = x$
- M_c_sepConv_(3 × 3) : $M_c - DR -$ Spatial_sepconv_(3 × 3) $-$ Conv(1 × 1) $- BN - M_c$
- M_c_sepConv_(5 × 5) : $M_c - DR -$ Spatial_sepconv_(5 × 5) $-$ Conv(1 × 1) $- BN - M_c$
- M_c_sepConv_(7 × 7) : $M_c - DR -$ Spatial_sepconv_(7 × 7) $-$ Conv(1 × 1) $- BN - M_c$, where DR is Dropout(0.5) and Relu, BN is batch normalization, M_c_sepConv_$(k \times k)(k = 3, 5, 7)$ are the designed multiscale convolution operators with channel attention, and Spatial_sepconv_$(k \times k)$ = Conv$(k \times 1)$ − Conv$(1 \times k)$. Specifically, M_c is expressed as

$$\begin{aligned} M_c(x) &= \sigma(\text{MLP}(\text{AvgPool}(x)) + \text{MLP}(\text{MaxPool}(x))) \\ &= \sigma\left(W_1\left(W_0\left(x_{\text{avg}}\right)\right) + W_1\left(W_0\left(x_{\text{max}}\right)\right)\right) \end{aligned} \tag{1}$$

where σ denotes the sigmoid function, $x \in \mathbb{R}^{C\times H\times W}$ is an intermediate feature for input, $W_0 \in \mathbb{R}^{C/r\times C}$, and $W_1 \in \mathbb{R}^{C\times C/r}$ $(r = 16)$. The output of spectral attention [50] M_c is defined as $M_c(x) \otimes x$, where $\otimes$ denotes elementwise multiplication.

2) Search Strategy: Once the search space is designed, the search for the cells can be performed, as illustrated in Fig. 2 (a detailed representation of part (a) from Fig. 1). The supernet consists of a reduction cell (Cell1) and a normal cell (Cell2), where all the operators adjacent to the input nodes have stride values of two and one, respectively. DARTS searches architecture by cells, which is represented as a directed acyclic graph composed of n-ordered intermediate nodes $\{n_0, n_1, \ldots, n_{n-1}\}$ and directed edges; each edge corresponds to a designed candidate operator (e.g. skip_connect, max pooling, and convolution) within the search space $O = \{o^{(i,j)} | 0 \le i < j \le h - 1\}$. The output of the cell C_l is expressed by formulas (2)–(4)

$$\overline{o}^{(i,j)}\left(n_j\right) = \sum_{o\in O} \frac{\exp\left(\alpha_o^{(i,j)}\right)}{\sum_{o'\in O}\exp\left(\alpha_{o'}^{(i,j)}\right)} o\left(n_j\right) \tag{2}$$

$$n_i = \sum_{j<i} \overline{o}^{(i,j)}\left(n_j\right) \tag{3}$$

$$C_l = \text{concat}(n_i) \tag{4}$$

where $\alpha_o^{(i,j)}$ is the coefficient of operator $o^{(i,j)}$ between node n_i and n_j. Then, the goal addressed by DARTS can be described as a two-layer optimization problem

$$\begin{aligned} \overset{*}{\alpha} &= \min_{\alpha} L_{\text{fsl}}^1\left(\overset{*}{\omega}, \alpha\right) \\ &\text{s.t. } \omega^*(\alpha) = \arg\min_{\omega} L_{\text{fsl}}^2(\omega, \alpha) \end{aligned} \tag{5}$$

where the architecture optimization and training of supernet are bilevel optimizing with $\alpha = \{\alpha_o^{(i,j)} | i, j., o \in O\}$ as the connection weight of operator and ω as the weight in supernet. The metric loss functions L_{fsl}^1 and L_{fsl}^2 will be introduced in detail in Section III-C. At the end of the search, the operator from n_i to n_j is determined by $\overline{o}^{(i,j)} = \arg\max_{o\in O}\alpha_o^{(i,j)}$.

B. FSL in Multisource and Target Domain

In HCFSL-NAS, an FSL framework is established in the multisource and target domains. Each dataset in the multisource domains is denoted as D_u, where u represents the source-domain index ranging from 1 to 2, and each dataset has c_u classes. During each episode, we randomly select c classes from each source dataset and sample k labeled samples per class to form a support set S. Similarly, n_s labeled samples are randomly sampled from the same c classes to form a query set Q, which excludes the samples in the support set. Thus, the support set in multisource domains is denoted as $S_u = \{(x_i, y_i)\}_{i=1}^{c\times k}$, and the query set is denoted as $Q_u = \{(x_j, y_j)\}_{j=1}^{c\times n_s}$. The target-domain dataset D_t is divided into the few-shot dataset with labeled samples and the test dataset with unlabeled samples. Similarly, a support set is $S_t = \{(x_i, y_i)\}_{i=1}^{c\times k}$ of k samples with c sampled classes for each episode and a query set $Q_t = \{(x_j, y_j)\}_{j=1}^{c\times n_t}$ of n_t samples with same classes.

1) Training in Multisource and Target: After the final-net is searched by the supernet in Section III-A, the training process depicted in Fig. 3 (a detailed representation of part (b) from Fig. 1) is conducted by the episode-based metatraining between the source and target datasets that facilitate collaboratively learning a shareable discriminant embedding space for both the source and target classes. First, three mapping layers, i.e., M_1, M_2, and M_t, are used to make the input dimension on multisource and target-domain datasets equal before fed into the final-net. The feature output of mapping layers is $F' = F \times M$, where $F \in \mathbb{R}^{H\times W\times ch}$ is the input data, $M \in \mathbb{R}^{ch\times 100}$, and ch is determined by the number of the input spectral bands. Then, the feature output of the final-net on different domains can be expressed as

$$\begin{cases} f_u^s := \text{Finalnet}(M_u(S_u)) \\ f_u^q := \text{Finalnet}(M_u(Q_u)) \end{cases} \tag{6}$$

where $u \in \{1, 2, t\}$ is determined by the domain dataset.

Generally, a metric loss function is used to perform the FSL for HSIC. In our model, the total loss function of multisource domains FSL with D_1 and D_2 datasets is given by

$$L_{\text{fsl}}^s = L_{\text{fsl}}^1 + L_{\text{fsl}}^2. \tag{7}$$

Similarly, the total loss function of target-domain FSL is given by

$$L_{\text{fsl}}^t = L_{ce}^t + L_{sp}^t. \tag{8}$$

The details of (7) and (8) will be described in detail in Section III-C.

2) Test in Target: During the testing phase in the target domain, the classification process involves a sequence of well-defined steps 1)–3), as depicted in Fig. 1(c).

1) Uniform dimensions of the target data x_i by employing the mapping layer M_t, i.e., $M_t(x_i)$.
2) Obtain the embedded features of the target data through Finalnet$(M_t(x_i))$.

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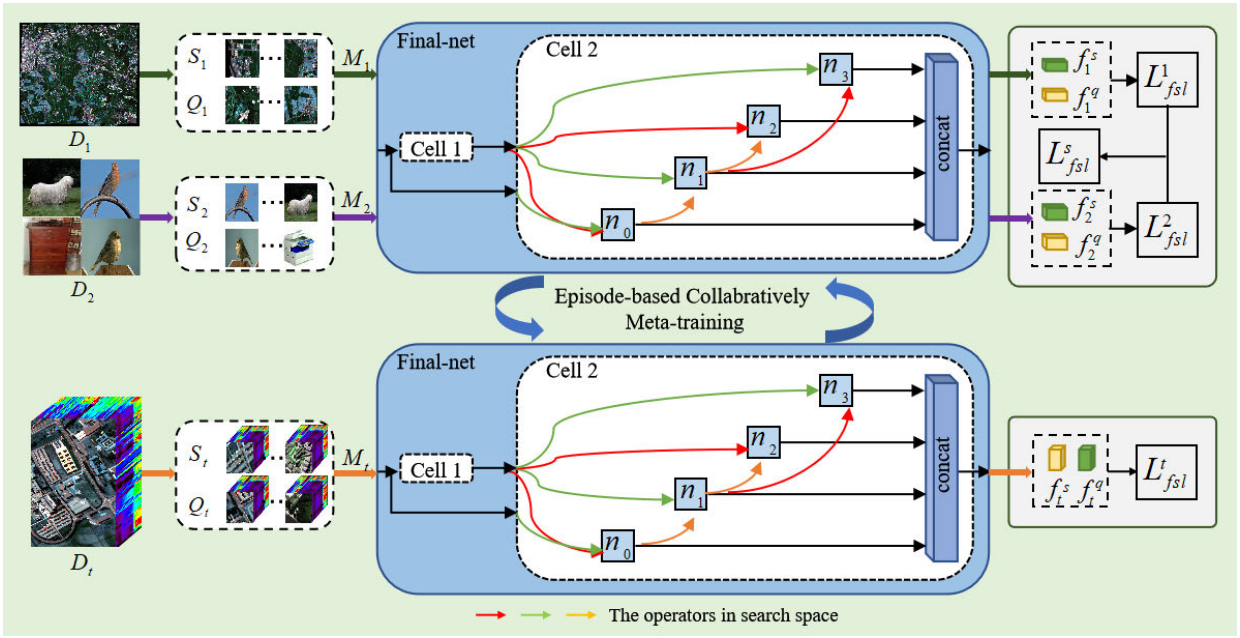


Fig. 3. Framework of the proposed HCFSL-NAS in the training phase.

3) Train the NN classifier by using the obtained embedded features with labels, and then, the unlabeled data are classified.

C. Metric Function

To mitigate the problem of misclassification caused by spectral variations within the same class and similarities across different classes in HSI, we propose a fusion loss function that integrates pointwise-based CE loss with pairwise-based adaptive sparse (SP) loss to enhance the interclass distance and diminish the intraclass distance. For individual samples, the pointwise-based CE loss conducts fine-gained classification training on each sample to enhance the aggregation and discrimination of samples. For group samples, the pairwise-based SP loss models the differences between pairs of samples with the same and different classes to promote interclass variations. By incorporating this fusion loss function, our objective is to enable the embedding extractor to acquire a metric space in which sample features exhibit small intraclass spacing and large interclass spacing.

In this article, we conduct FSL in the multisource and target domains. The fusion loss designs for the domain datasets are given by

$$L^u_{\text{fsl}} = L^u_{ce} + L^u_{sp}, u \in \{1, 2, t\}. \tag{9}$$

According to S_u and Q_u from D_u, the pointwise-based CE loss L^u_{ce} can be obtained by (10), and the pairwise-based SP loss [51] L^u_{sp} can be defined by (12)

$$L^u_{ce} = E_{S_u,Q_u}[-\sum_{(x,y)\in Q_u} \log p(y = i|x)] \tag{10}$$

$$p(y_j = i|x_j \in Q_u) = \frac{\exp(-d(f^q_{uj}, f^s_{ui}))}{\sum_{i=1}^{C} \exp(-d(f^q_{uj}, f^s_{ui}))} \tag{11}$$

where $d(\cdot)$ is the Euclidean distance function, f^q_{uj} is the embedded features f^q_u of the jth samples in query set $Q_u (u \in \{1, 2, t\})$, f^s_{ui} is the embedded features f^s_u of the ith class in support set S_u, c is the number of classes in each task, and y_j is the label of the sample x_j in Q_u

$$L^u_{sp} = \frac{1}{c}\sum_{j}^{c} \log\left(1 + e^{\frac{s^-_j - (\alpha_j s^+_{j,h} + (1-\alpha_j) s^+_{j,lh})}{\tau}}\right) \tag{12}$$

where $s^-_j \approx \tau \log(\sum_{n=1}^{N}\sum_{r=1,r\neq j}^{c}\sum_{m=1}^{N} e^{z^j_n z^r_m/\tau})$ is the soft version of the similarity of the hardest negative pair of the ith class, $s^+_{j,h} \approx -\tau \log(\sum_{n=1}^{N}\sum_{m=1}^{N} e^{-z^j_n z^r_m/\tau})$ is the similarity of the hardest positive pair in the ith class, $s^+_{j,lh} = \tau \log(\sum_{n=1}^{N} 1/\sum_{m=1}^{N} e^{-z^j_n z^r_m/\tau})$ is the similarity of the least-hardest positive pair in the ith class, α_j is the harmonic mean of $s^+_{j,lh}$ and $s^+_{j,h}$, z^j_n is the normalized embedding vector $w^j_n = -d(f^q_{uj}, f^s_{ui})$, τ is a temperature parameter, and N is the number of samples in a minibatch.

IV. Experiment and Analysis

In this section, four real hyperspectral image datasets are used to evaluate the proposed HCFSL-NAS method. First, we introduce the datasets used in our experiments and implementation details in Sections IV-A and IV-B, respectively. Second, we provide a detailed ablation analysis of the effectiveness of the searched architecture and the designed multisource domains in Section IV-C. Third, we compare the performance of HCFSL-NAS with other methods using qualitative metrics, including overall accuracy (OA), average accuracy (AA), and the kappa coefficient (KP) in Section IV-D. Furthermore, to demonstrate the complexity of HCFSL-NAS, we present the comparison results in terms of training parameters (Params), FLOPs, GPU memory usage (GPU), and the test time (TST) in Section IV-E. Finally,

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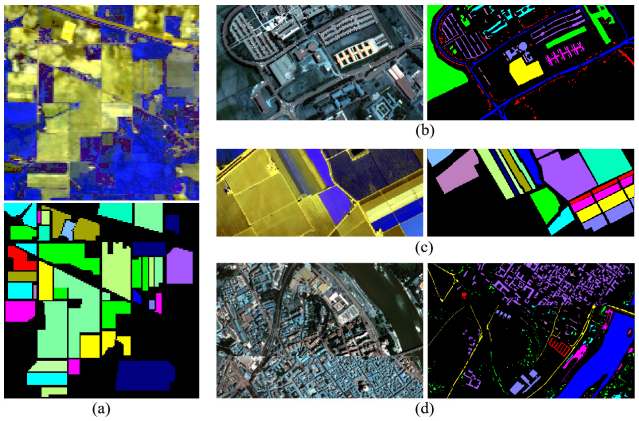


Fig. 4. False color composites and ground-truth maps of target datasets. (a) IP dataset. (b) PU dataset. (c) SA dataset. (d) PC dataset.

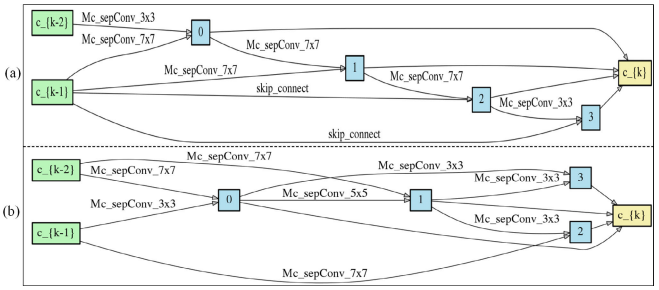


Fig. 5. Optimal structure of cells based on the HCFSL-NAS on source datasets. (a) Cell1. (b) Cell2.

we provide a detailed analysis of the robustness and model regularization in Section IV-F.

A. Datasets

The experiment datasets used in this work consist of multisource-domain datasets with sufficient labeled samples and target-domain datasets with few labeled samples. For the multisource datasets, as the research in [26], [27], and [28], we select the mini-ImageNet dataset (MI) as D_1 and the Chikusei dataset (CK) as D_2, which together form the multisource-domain datasets. In terms of target datasets, we follow the approach in [24], [25], and [27] by conducting comparative experiments on four representative HSI datasets: Indian Pines (IP), Pavia University (PU), Salinas (SA), and Pavia Center (PC). The details of the target datasets[1] are listed in Table II with their false color composites and ground-truth maps shown in Fig. 4.

1) Source-Domain Dataset: MI, which is a subset of the ImageNet Large Scale Visual Recognition Challenge (ILSVRC), consists of 100 categories with 600 annotated images per category. CK was captured in Chikusei, Japan, in 2014 using the Headwall Hyperspec-VNIR-C sensor. It includes 128 spectral bands ranging from 343 to 1018 nm with 2517 × 2335 pixels and a spatial resolution of 2.5 m comprising 19 classes of objects.

[1]https://www.ehu.eus/ccwintco/index.php/Hyperspectral_Remote_Sensing_Scenes

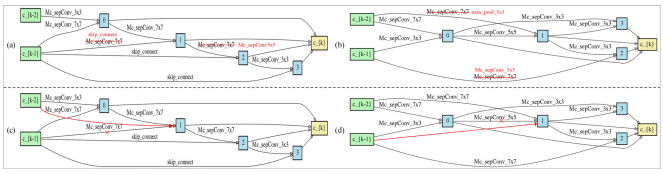


Fig. 6. Architecture for cells. (a) HCFSL-NAS_Rop cell1. (b) HCFSL-NAS_Rop cell2. (c) HCFSL-NAS_Rtop cell1. (d) HCFSL-NAS_Rtop cell2.

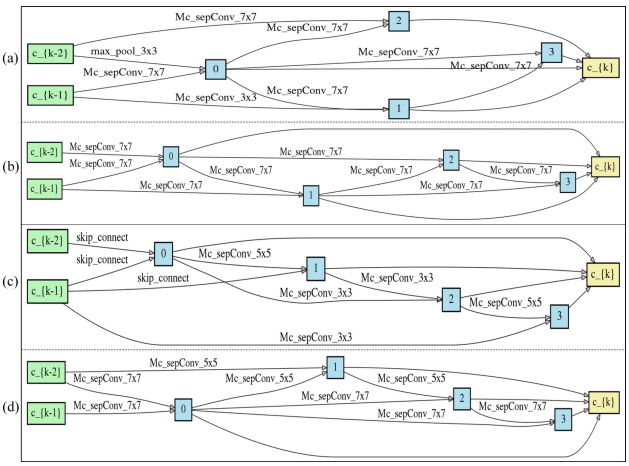


Fig. 7. Optimal structure of cells based on the HFSL-NAS on a single-source dataset. (a) Cell1 of MI. (b) Cell2 of MI. (c) Cell1 of CK. (d) Cell2 of CK.

2) Target-Domain Dataset: IP was captured by the AVIRIS sensor in Indiana. It has 145 × 145 pixels with 200 spectral bands and a spatial resolution of 20 m consisting of 16 crop categories. SA was captured by the AVIRIS sensor in SA Valley, California. It has a spatial resolution of 3.7 m and 512 × 217 pixels, which contains 204 spectral bands and 16 crop categories. PU and PC were collected during a flight campaign over the Pavia region in Northern Italy using the reflective optics system imaging spectrometer (ROSIS) sensor in 2001, both of which have a spatial resolution of 1.3 m with 610 × 340 and 1096 × 715 pixels, respectively. PU ad PC have 103 and 102 spectral bands, respectively, including nine classes of ground objects, as listed in Table II.

B. Implementation Details

To ensure fairness and comparability in the experiments, we maintain consistency across all comparative experiments. Specifically, we randomly choose five labeled samples per class from the target domain in single-domain methods, such as A^2S^2K-RNet, LMSS-NAS, and S3Net. For multidomain methods, an additional 200 labeled source-domain samples are randomly selected, such as DFSL, DCFSL, HFSL, CMFSL, CA-CFSL, and HCFSL-NAS. The remaining samples from the target domain are reserved for testing.

For each episode, the input size of the cube $\mathbb{R}^{H\times W\times ch}$ is set to 9 × 9 × bands in CK and target datasets, following [24], [25], [27], and [26] to maintain the same size as the spatial characteristics of the hyperspectral data in cross-domain scenarios. In addition, in MI, the input size is set to 33 × 33 × 3 per image, followed by [28] to leverage its

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TABLE II

ATTRIBUTE ANALYSIS OF FOUR TARGET HYPERSPECTRAL DATASETS

Data Set	IP			SA			PU			PC		
No.	Category	Labelled samples	Color	Category	Labelled samples	Color	Category	Labelled samples	Color	Category	Labelled samples	Color
1	Alfalfa	46		Brocoli_green_weeds_1	2009		Asphalt	6631		Water	65971	
2	Corn-notill	1428		Brocoli_green_weeds_2	3726		Meadows	18649		Trees	7598	
3	Corn-mintill	830		Fallow	1976		Gravel	2099		Asphalt	3090	
4	Corn	237		Fallow_rough_plow	1394		Trees	3064		Self-Blocking Bricks	2685	
5	Grass-pasture	483		Fallow_smooth	2678		Painted metal sheets	1345		Bitumen	6584	
6	Grass-tree	730		Stubble	3959		Bare Soil	5029		Tiles	9248	
7	Grass-pasture-mowed	28		Celery	3579		Bitumen	1330		Shadows	7287	
8	Hay-windrowed	478		Grapes_untrained	11271		Self-Blocking Bricks	3682		Meadows	42826	
9	Oats	20		Soil_vinyard_develop	6203		Shadows	947		Bare Soil	2863	
10	Soybean-notill	972		Corn_senesced_green_weeds	3278							
11	Soybean-minitill	2455		Lettuce_romaine_4wk	1068							
12	Soybean-clean	593		Lettuce_romaine_5wk	1927							
13	Wheat	205		Lettuce_romaine_6wk	916							
14	Woods	1265		Lettuce_romaine_7wk	1070							
15	Buildings-Grass-Trees-Drives	386		Vinyard_untrained	7268							
16	Stone-Steel-Towers	93		Vinyard_vertical_trelis	1807							
	Total	10249		Total	54129		Total	42776		Total	148152	

TABLE III

EFFECTIVENESS OF HCFSL-NAS (%)

	D_t	IP	PU	SA	PC
HCFSL-NAS	OA	77.16±1.46	88.51±1.34	92.40±0.51	98.01±0.39
	AA	84.98±1.25	88.73±1.74	95.78±0.56	94.77±0.75
	KP	74.15±1.55	84.97±1.63	91.55±0.56	97.19±0.56
	D_t	IP	PU	SA	PC
HCFSL-NAS_Rop	OA	74.75±2.50	87.95±2.12	92.18±1.43	97.86±0.47
	AA	84.91±1.62	87.47±2.76	95.52±1.30	94.41±0.93
	KP	71.66±2.66	84.17±2.72	91.31±1.59	96.98±0.66
	D_t	IP	PU	SA	PC
HCFSL-NAS_Rtop	OA	76.64±2.84	87.03±3.48	91.43±1.10	97.96±0.42
	AA	85.08±1.53	86.39±1.99	95.35±0.84	94.99±0.91
	KP	73.64±3.07	83.04±4.34	90.48±1.22	97.11±0.59

TABLE IV

EXPERIMENTAL RESULTS (%) OF HCFSL-NAS ON FOUR DATASETS FOR MULTISOURCE LEARNING

source set	target set	IP	PU	SA	PC
MI	OA	72.64±2.48	84.24±3.84	90.32±1.94	97.50±0.51
	AA	82.55±1.07	85.35±3.69	94.58±1.30	94.29±0.97
	KP	69.10±2.60	79.54±4.76	89.26±2.13	96.47±0.73
source set	target set	IP	PU	SA	PC
CK	OA	76.68±4.19	84.96±2.06	90.18±1.76	97.38±0.73
	AA	**85.25±1.87**	86.63±2.82	94.36±1.39	94.16±1.15
	KP	73.64±4.57	80.43±2.62	89.09±1.93	96.29±1.02
source set	target set	IP	PU	SA	PC
MI&CK	OA	**77.16±1.39**	**88.51±1.27**	**92.40±0.48**	**98.01±0.37**
	AA	84.98±1.18	**88.73±1.65**	**95.78±0.53**	**94.77±0.71**
	KP	**74.15±1.47**	**84.97±1.55**	**91.55±0.54**	**97.19±0.53**

rich spatial and texture information. Specifically, an FSL task in each episode is c-ways k-shot n-query, where c, k, and n represent the number of selected classes, labeled samples per class in support set, and labeled samples per class in query set, respectively. In general, c is set to the number of classes in the target dataset (e.g., $c_{pu} = c_{pc} = 9$, $c_{sa} = c_{ip} = 16$). According to previous FSL methods for each task ($k = 1$, $n = 19$) with a single-source dataset and a target dataset, we ensure the total number in source-domain datasets, i.e., $k = 1$, $n_s = 9$. For the target dataset, k and n_t are the same as previous FSL methods. In order to ensure the stability of comparison, each experiment is repeated ten times, and the average is taken as the final result. The number of the searching iteration is set to 500, and the training iteration is set to 20 000. The Adam optimizer is used to optimize the network with the default weight decay = 0.0003, the drop path prob = 0.2, and the learning rate = 0.004. After the searching iteration, the optimal architectures of Cell1 and Cell2 in the source domain are obtained, as shown in Fig. 5.

Regarding the model sampling configuration in the comparative methods, a uniform overlap sampling mode is employed to calculate the pixels in the specified window of each labeled sample. More precisely, the window sizes of LMSS-NAS, S3Net, and HFSL are set to 27 × 27, {13 × 13, and 7 × 7}, and 33 × 33, respectively, while A^2S^2K-RNet, DFSL, DCFSL, CMFSL, and CA-CFSL are set to 9 × 9. Notably, CMFSL selects 100 spectral bands for all datasets using the OCBS [52] method, while DFSL employs the GRBS [53] method to choose 100 bands for source datasets. In the case of S3Net, the number of bands is set to 60 for IP and SA and 30 for PU and PC through the PCA method. The hyperparameters of the comparison methods are set according to the corresponding articles. All experiments are implemented on a single GTX 1080 Ti GPU, which has 11-GB memory.

C. Ablation Study

1) Effectiveness of the Searched Architecture: In this article, the cells searched by HCFSL-NAS are composed of a combination of candidate operations in the search space. By searching for the optimal operation combination in the cell, we can effectively learn data features adaptively and improve classification accuracy. To verify whether HCFSL-NAS enhances accuracy through the exploration of efficient architectures or merely integrates various branching and convolution operators, we conducted manual modifications to the architecture searched by HCFSL-NAS in two distinct ways. As shown in Fig. 6, the modified operators in normal and reduction cells are highlighted in red. The two types of modifications include: 1) randomly replacing two selected operators with others in the cell (termed as HCFSL-NAS_Rop) and 2) changing the connection relationships between the nodes inside each cell (termed as HCFSL-NAS_Rtop). From Table III, it can be seen that the performance of the manually modified structures is inferior to that of the automatically searched architecture, which shows the excellent performance of NAS for the automatic design of model architecture.

2) Effectiveness of Multisource Domains: To verify the effectiveness of multisource source-domain learning, this study conducted ablation experiments to compare single-task source-domain learning with multisource source-domain learning. In single-task source-domain learning, separate feature extractors were designed for each source domain. Fig. 7 shows the

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TABLE V

CLASSIFICATION RESULTS (%) ON THE IP DATASET (FIVE LABELED SAMPLES PER CLASS)

Models	A^2S^2K-RNet [42] 2021TGRS	LMSS-NAS [37] 2023TGRS	S3Net [43] 2022TGRS	DFSL [24] 2019TGRS	DCFSL [27] 2022TGRS	HFSL [28] 2022LGRS	CMFSL [25] 2022TIP	CA-CFSL [26] 2023TGRS	HCFSL-NAS
1	37.52±19.67	18.43±3.64	100.00±0.00	92.20±7.16	94.88±7.58	99.51±1.03	96.34±6.83	97.80±3.53	99.27±1.18
2	71.20±12.94	64.05±6.94	53.15±12.49	34.76±10.89	39.75±12.35	55.00±10.11	49.13±9.59	45.97±12.32	65.36±11.91
3	47.87±17.52	65.25±5.03	48.61±20.39	37.96±8.86	47.44±5.90	59.21±16.01	53.78±7.68	51.89±11.80	60.76±10.65
4	42.29±12.17	70.05±12.34	85.17±22.53	67.41±15.41	68.75±17.54	93.79±7.11	89.61±7.92	75.09±20.80	83.66±17.29
5	78.67±16.38	88.27±6.17	79.08±11.39	60.88±10.47	68.22±8.47	72.76±12.63	72.76±9.56	76.03±7.19	80.63±9.72
6	85.12±12.27	75.66±5.94	96.34±3.02	77.12±9.77	83.06±7.81	70.84±12.97	88.76±5.78	90.61±4.37	92.06±6.29
7	27.81±10.72	38.28±7.05	100.00±0.00	97.83±4.23	97.83±2.29	100.00±0.00	100.00±0.00	98.26±3.04	98.26±5.50
8	94.82±6.11	89.37±3.51	88.92±19.97	89.60±4.35	83.51±13.82	99.83±0.47	91.46±7.79	86.11±13.72	94.12±9.39
9	14.11±6.50	6.60±1.65	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00
10	53.62±11.48	40.83±4.04	71.45±16.68	49.30±11.43	57.57±8.16	65.42±7.34	57.58±8.49	61.40±10.04	76.94±9.23
11	81.58±7.45	87.21±7.37	43.55±32.38	45.79±12.16	54.27±12.69	70.91±11.71	51.69±16.22	61.14±13.51	71.42±11.67
12	52.23±10.68	71.86±13.70	55.02±25.32	37.18±8.35	41.60±7.86	67.53±18.73	52.16±12.54	42.16±13.48	70.54±13.32
13	77.90±9.11	38.50±4.82	98.85±1.75	96.05±7.95	98.85±1.25	99.45±0.76	99.10±0.74	99.90±0.21	97.35±2.57
14	96.02±2.43	91.20±2.11	93.87±9.15	76.21±12.42	83.98±11.04	91.94±5.81	82.33±18.60	86.02±9.48	91.32±5.19
15	51.18±8.42	71.79±3.74	86.61±11.91	73.70±8.66	64.23±8.90	92.28±10.17	82.83±10.25	71.68±9.67	79.84±10.46
16	48.97±17.42	38.52±5.81	99.32±0.96	97.73±3.75	98.98±1.56	93.30±6.66	99.66±0.77	99.43±1.23	98.18±3.48
OA	64.66±3.05	64.92±1.21	66.97±9.10	55.56±2.53	61.31±2.02	73.35±3.18	65.11±3.82	66.34±3.11	**77.16±1.46**
AA	60.06±3.16	59.74±1.36	81.25±5.13	70.86±1.27	73.93±1.37	83.24±2.65	79.20±1.52	77.72±2.16	**84.98±1.25**
KP	60.99±3.14	61.22±1.30	63.42±9.65	50.29±2.57	56.63±2.01	69.86±3.55	60.91±3.95	62.06±3.33	**74.15±1.55**

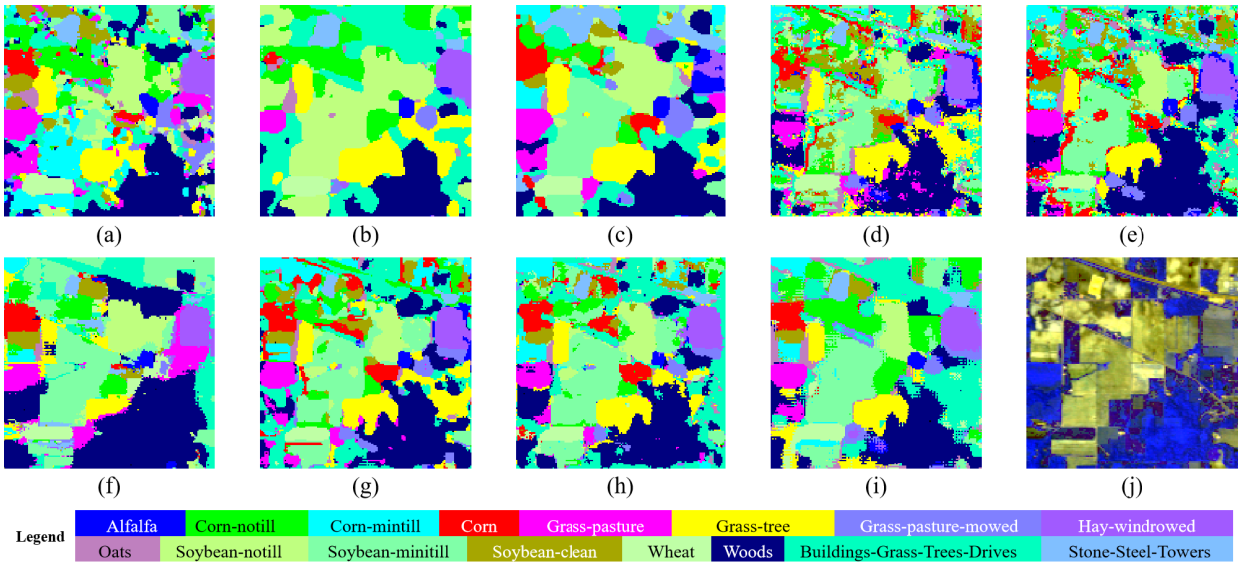


Fig. 8. Full classification maps obtained via various algorithms for the IP dataset. (a) A^2S^2K-RNet. (b) LMSS-NAS. (c) S3Net. (d) DFSL. (e) DCFSL. (f) HFSL. (g) CMFSL. (h) CA-CFSL. (i) HCFSL-NAS. (j) False-color image.

TABLE VI

CLASSIFICATION RESULTS (%) ON THE PU DATASET (FIVE LABELED SAMPLES PER CLASS)

Models	A^2S^2K-RNet [42] 2021TGRS	LMSS-NAS [37] 2023TGRS	S3Net [43] 2022TGRS	DFSL [24] 2019TGRS	DCFSL [27] 2022TGRS	HFSL [28] 2022LGRS	CMFSL [25] 2022TIP	CA-CFSL [26] 2023TGRS	HCFSL-NAS
1	67.22±13.76	91.11±1.96	79.72±11.17	63.22±16.15	77.79±9.43	74.56±10.90	76.80±13.18	79.64±13.93	87.76±6.80
2	92.23±4.39	91.60±1.48	76.98±15.88	82.23±7.70	81.41±10.96	89.06±6.28	82.86±11.25	85.82±6.37	90.00±4.84
3	39.18±9.36	54.34±5.88	74.74±12.47	58.79±9.84	65.25±11.45	81.19±10.14	67.07±14.15	65.80±10.57	67.50±15.16
4	69.71±16.52	36.09±3.34	89.45±6.91	91.32±3.72	92.74±2.51	94.52±4.47	93.27±3.60	93.06±5.28	91.15±4.12
5	95.36±4.28	90.44±0.37	99.93±0.17	99.51±0.60	99.34±0.91	99.87±0.15	99.43±1.17	99.67±0.31	98.92±1.41
6	52.12±9.61	56.80±2.31	76.62±17.40	56.19±14.06	70.84±11.58	88.79±13.20	87.44±9.07	78.34±11.10	86.24±7.85
7	45.80±13.59	79.49±6.63	99.71±0.40	79.83±7.53	76.81±9.56	90.20±9.38	82.63±7.36	82.39±9.05	93.28±3.64
8	47.85±13.06	84.26±2.92	83.42±17.10	56.63±13.20	61.74±13.19	96.52±3.99	73.83±11.52	64.95±9.94	87.70±7.04
9	49.54±20.50	45.81±2.70	89.94±9.39	99.29±0.57	98.03±3.63	97.55±5.71	99.34±0.81	99.51±0.71	96.00±5.89
OA	63.77±4.80	70.92±1.87	80.41±6.38	74.37±3.88	78.72±4.34	87.99±2.73	82.53±4.46	82.36±2.97	**88.51±1.34**
AA	62.11±3.23	69.99±0.79	85.61±3.65	76.34±2.26	80.44±2.06	**90.25±2.03**	84.74±1.41	83.24±2.05	88.73±1.74
KP	55.42±5.06	63.62±2.13	75.18±7.07	66.89±4.57	72.67±4.80	84.40±3.42	77.68±5.13	77.17±3.50	**84.97±1.63**

architecture of the cells obtained when treating MI and CK as single-task source domains. It can be observed that MI tends to choose large-scale convolutions, displaying a relatively uniform scale selection and lacking skip connections. In contrast, CK tends to select convolutions of medium and small scales, exhibiting more diverse choices and emphasizing a deeper network compared to MI.

Table IV presents the classification results of the ablation experiments. It can be observed that when MI and CK are treated as independent single-source domains, the MI domain

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TABLE VII
CLASSIFICATION RESULTS (%) ON THE SA DATASET (FIVE LABELED SAMPLES PER CLASS)

Models	A^2S^2K-RNet [42] 2021TGRS	LMSS-NAS [37] 2023TGRS	S3Net [43] 2022TGRS	DFSL [24] 2019TGRS	DCFSL [27] 2022TGRS	HFSL [28] 2022LGRS	CMFSL [25] 2022TIP	CA-CFSL [26] 2023TGRS	HCFSL-NAS
1	98.57±3.16	71.94±3.21	99.72±0.75	97.78±2.40	99.03±0.88	100.00±0.00	99.55±0.85	99.68±0.86	98.77±1.34
2	99.35±1.22	97.54±1.91	98.04±3.46	99.10±2.30	99.87±0.11	96.09±4.31	99.87±0.34	99.95±0.11	99.85±0.28
3	94.23±8.63	99.75±0.19	99.86±0.42	93.55±6.31	93.43±6.23	99.25±0.96	97.97±2.60	92.21±9.18	98.12±4.14
4	93.45±4.95	77.47±2.88	79.81±42.06	99.45±0.63	99.27±0.74	99.71±0.35	99.50±0.63	99.77±0.20	98.58±1.16
5	98.16±1.65	99.43±0.77	96.70±2.69	91.49±3.36	91.72±3.09	95.31±3.94	92.41±2.81	91.79±2.92	96.59±1.75
6	98.96±1.73	88.05±1.17	99.06±1.35	99.27±0.60	99.41±0.65	99.31±1.28	97.42±2.02	99.54±0.59	99.77±0.31
7	99.54±0.59	100.00±0.00	99.85±0.28	99.57±0.38	99.15±0.96	99.13±1.40	99.58±0.67	99.21±1.77	99.66±0.46
8	84.80±5.29	79.38±3.56	83.17±8.30	71.68±13.84	73.71±12.08	70.66±13.59	71.16±11.12	75.27±9.06	80.93±4.37
9	99.41±0.34	99.95±0.10	99.95±0.07	93.81±3.58	99.78±0.21	99.49±1.00	96.63±3.32	99.69±0.63	99.78±0.30
10	96.12±2.67	94.76±1.96	92.70±3.56	78.66±13.70	85.10±5.87	94.88±9.28	83.66±7.70	87.78±6.94	89.21±5.01
11	83.41±22.52	97.76±1.01	89.51±31.46	95.23±5.21	98.61±1.45	99.41±0.68	99.18±0.86	99.26±0.93	97.81±4.02
12	97.02±3.39	99.70±0.27	93.83±8.25	99.17±0.87	98.64±1.67	96.29±3.16	99.70±0.65	99.78±0.41	99.48±0.67
13	97.82±2.33	94.80±1.57	67.38±47.07	99.28±1.77	98.92±0.67	98.53±1.63	98.52±1.86	99.08±0.73	99.00±1.23
14	82.75±17.69	92.86±3.77	92.63±7.25	98.37±1.03	98.52±1.78	97.87±1.71	99.46±0.48	98.54±0.87	97.37±2.39
15	59.74±9.95	66.53±2.12	80.18±29.79	71.77±10.48	74.22±7.51	76.14±13.24	76.69±7.94	74.25±8.05	83.12±6.30
16	97.15±7.54	99.94±0.16	87.43±30.79	89.53±7.72	89.11±6.99	92.10±10.38	94.33±4.46	91.42±4.12	94.48±5.04
OA	86.82±2.44	87.21±0.95	90.93±3.66	86.89±1.73	88.83±2.09	89.19±2.90	88.52±2.03	89.46±1.69	**92.40±0.51**
AA	92.53±1.80	91.24±0.74	91.24±4.84	92.36±0.84	93.66±1.00	94.64±1.57	94.10±0.77	94.20±0.95	**95.78±0.56**
KP	85.41±2.69	85.77±1.06	89.93±4.02	85.45±1.89	87.60±2.29	88.01±3.19	87.27±2.23	88.29±1.86	**91.55±0.56**

TABLE VIII
CLASSIFICATION RESULTS (%) ON THE PC DATASET (FIVE LABELED SAMPLES PER CLASS)

Models	A^2S^2K-RNet [42] 2021TGRS	LMSS-NAS [37] 2023TGRS	S3Net [43] 2022TGRS	DFSL [24] 2019TGRS	DCFSL [27] 2022TGRS	HFSL [28] 2022LGRS	CMFSL [25] 2022TIP	CA-CFSL [26] 2023TGRS	HCFSL-NAS
1	99.81±0.14	99.75±0.12	98.76±1.15	99.86±0.13	99.79±0.19	99.45±0.28	99.57±0.56	99.80±0.19	99.78±0.19
2	92.37±6.17	74.67±0.70	91.06±6.26	92.19±3.19	90.39±4.04	89.88±7.98	90.15±4.15	91.30±3.70	94.26±1.38
3	66.80±6.31	61.55±4.28	85.36±14.82	93.88±3.13	93.12±5.37	77.26±11.73	92.77±3.18	94.21±2.95	92.70±3.74
4	56.99±8.46	75.24±12.57	96.59±5.82	86.71±13.32	94.24±5.22	93.92±8.88	94.09±6.51	93.61±5.58	91.23±6.42
5	86.05±6.84	97.90±0.82	86.13±6.24	77.45±10.81	90.40±5.19	96.61±3.40	92.47±4.77	91.72±3.89	93.86±6.05
6	83.11±6.43	87.46±1.53	98.14±1.69	86.49±9.12	95.07±3.62	82.05±10.92	95.49±3.27	97.18±1.88	97.71±1.92
7	92.80±6.58	99.72±0.10	89.66±5.80	82.82±4.71	83.81±2.36	86.89±8.04	84.69±4.53	85.49±2.63	86.51±3.33
8	99.28±0.78	96.77±0.54	97.86±2.17	88.11±6.00	98.38±1.21	97.74±1.43	98.48±0.60	98.84±0.67	99.46±0.47
9	60.12±12.46	71.53±3.56	92.80±8.58	98.72±1.66	98.04±1.49	98.14±3.46	99.14±0.94	98.07±1.59	97.39±1.86
OA	93.90±1.24	94.93±0.51	96.62±1.04	93.02±1.84	97.13±0.55	96.05±1.02	97.22±0.60	97.60±0.31	**98.01±0.39**
AA	81.98±2.53	84.95±1.38	92.93±2.36	89.58±1.65	93.69±0.88	91.33±1.58	94.09±0.84	94.47±0.51	**94.77±0.75**
KP	91.42±1.73	92.79±0.73	95.24±1.46	90.25±2.50	95.94±0.77	94.42±1.43	96.08±0.84	96.60±0.44	**97.19±0.56**

achieves better classification performance on SA and PC, while slightly lower performance is observed on IP and PU. This indicates that different source domains have their own advantages in single-task source-domain learning. However, when MI and CK are integrated into multisource learning, the learned classification knowledge significantly improves the accuracy and generalization of HSIC under limited labeled samples, achieving the best results across all four target datasets. This further demonstrates the effectiveness of multi-source source-domain learning in HSIC.

D. Classification Results

To evaluate the effectiveness of HCFSL-NAS, we compare with eight state-of-the-art methods, where the results of the comparison methods are derived from official code, containing DL-based methods (A^2S^2K-RNet[2] [42] and LMSS-NAS[3] [37] and FSL-based methods (S3Net[4] [43], DFSL[5] [24], DCFSL[6] [27], HFSL[7] [28], CMFSL[8] [25], and CA-CFSL [26]).

[2]https://github.com/suvojit-0 × 55aa/A2S2K-ResNet
[3]https://github.com/xh-captain/LMSS-NAS
[4]https://github.com/ZhaohuiXue/S3Net
[5]https://github.com/liubing220524/DFSL-for-hyperspectral-image-classification
[6]https://github.com/Li-ZK/DCFSL-2021
[7]https://github.com/Li-ZK/HFSL
[8]https://github.com/B-Xi/TIP_2022_CMFSL

The average results for OA, AA, and KP on IP, PU, SA, and PC are presented in Tables V–VIII. Three key observations can be derived from the quantitative classification results.

1) *Comparative Analysis With Non-FSL Methods:* Among the compared methods, the non-FSL methods A^2S^2K-RNet (OA: 64.66% in IP, 63.77% in PU, 86.82% in SA, and 93.90% in PC) performed worst on four target datasets. When considering only limited labeled data from the target domain for training, the advantage of DL is insufficient. While NAS-based LMSS-NAS (OA: 64.92% in IP, 70.92% in PU, 87.21% in SA, and 94.93% in PC) improved performance over A^2S^2K-RNet and achieved similar OA with traditional FSL-based DFSL (OA: 55.56% in IP, 74.37% in PU, 86.89% in SA, and 93.02% in PC), it lags behind other recent FSL methods due to overfitting caused by the use of limited labeled data.
2) *Comparative Analysis With the Handcrafted FSL Methods:* DCFSL (OA: 88.83% on SA) and S3Net (OA: 90.93% on SA) achieved similar OA and improved by at least 1.94% over DFSL on four datasets. The superiority of DCFSL lies in its ability to align the domains. Both CA-CFSL and CMFSL utilized attention mechanisms to further improve the OA by 1.95% and 2.21% over S3Net on PU. Benefiting from the selection of the natural image

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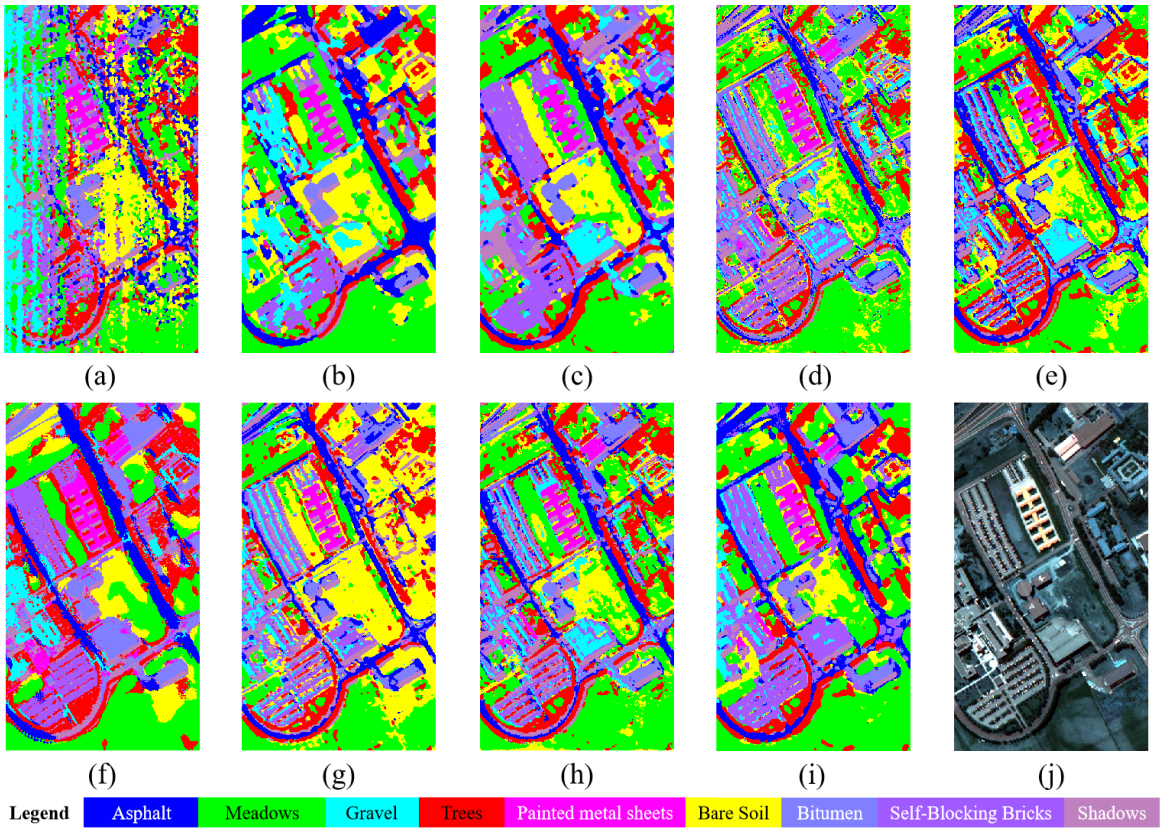


Fig. 9. Full classification maps obtained via various algorithms for PU dataset. (a) A^2S^2K-RNet. (b) LMSS-NAS. (c) S3Net. (d) DFSL. (e) DCFSL. (f) HFSL. (g) CMFSL. (h) CA-CFSL. (i) HCFSL-NAS. (j) False-color image.

dataset as the source domain, HFSL (OA: 73.75% on IP and 87.99% on PU) achieved better classification on the complex target datasets IP and PU compared to CA-CFSL, while S3Net and CA-CFSL improved OA by 1.74% and 1.55% over HFSL on SA and PC, respectively. This demonstrates that handcrafted FSL-based methods have their own advantages in various scenarios.

3) *Analysis of the Proposed HCFSL-NAS:* The proposed HCFSL-NAS shows the most competitive performance on four target datasets by leveraging its autodesigned feature extractor and the extensive knowledge learned from the multisource domains. For instance, it achieves significant improvements in OA of at least 3.81%, 0.52%, 1.47%, and 0.41% across IP, PU, SA, and PC datasets compared to other methods. Furthermore, the AA of HCFSL-NAS rises by 1.74 %, 1.14%, and 0.41% than that of suboptimal HFSL on IP, SA, and PC datasets, respectively. In addition, HCFSL-NAS exhibits the highest OA on PU compared with HFSL. The AA underperforms since the HFSL achieved the highest accuracy across six individual classes while the HCFSL-NAS was with its highest accuracies in three specific classes (Asphalt, Meadows, and Bitumen). Despite the lower AA, the number of labeled samples in the three classes where the HCFSL-NAS has an advantage covering 62% of the total labeled samples to obtain the highest KP, which is 4.29% higher than that of HFSL on IP.

In addition, for an intuitive comparison of the classification results among all methods, the corresponding full classification maps on the four novel datasets are displayed in Figs. 8–11, and we give the labeled classification maps in Figs. 1–4 in the Supplementary Material. Consistent with the qualitative results, we observe that as OA increases, the difference between the classification map and the false-color image shows a decreasing trend. Visually, the full classification maps of the proposed model on four novel datasets perform the best among all the methods. Thus, the combination of these quantitative observations with the quantitative results in Tables V–VIII confirms the effectiveness of our method. In addition, to further demonstrate the representativeness of our research, we explored novel datasets, including WHU-Hi-LongKou [54], WHU-Hi-HanChuan[9] [55], and the Botswana dataset, which better reflect real-world scenarios. More detailed experimental results can be seen in Tables I–III and Figs. 5–7 in the Supplementary Material.

E. Model Complexity

To analyze the computational complexity among different methods, we compare the number of parameters (Params), FLOPs, GPU memory usage (GPU), and the TST for the proposed HCFSL-NAS with other methods in the case of five labeled samples per class in the target domain. The numerical results on the four HSI datasets are presented in Table IX.

From the comparison results, we observe that, for different datasets, methods such as A^2S^2K-RNet, DCFSL, and HFSL exhibit a large number of Params due to the dense

[9]http://rsidea.whu.edu.cn/resource_WHUHi_sharing.htm

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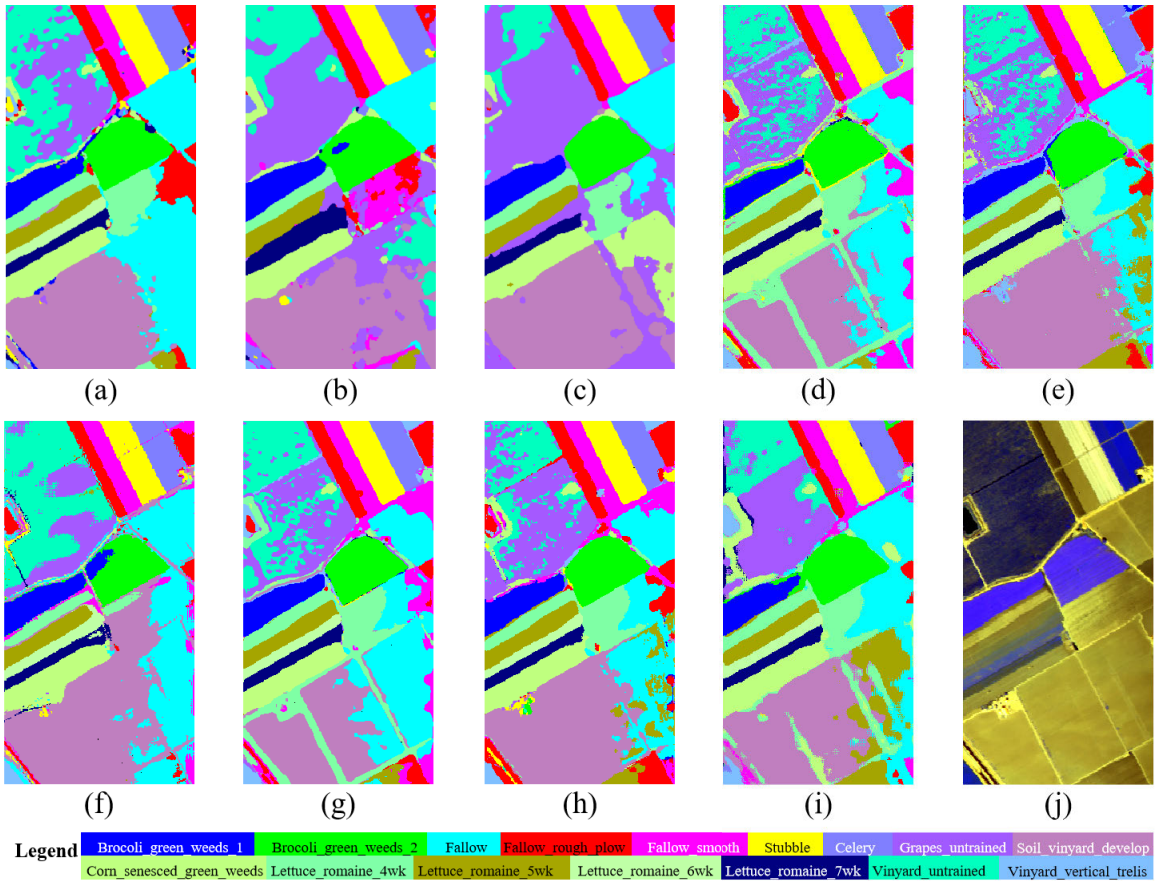


Fig. 10. Full classification maps obtained via various algorithms for the SA dataset. (a) A^2S^2K-RNet. (b) LMSS-NAS. (c) S3Net. (d) DFSL. (e) DCFSL. (f) HFSL. (g) CMFSL. (h) CA-CFSL. (i) HCFSL-NAS. (j) False-color image.

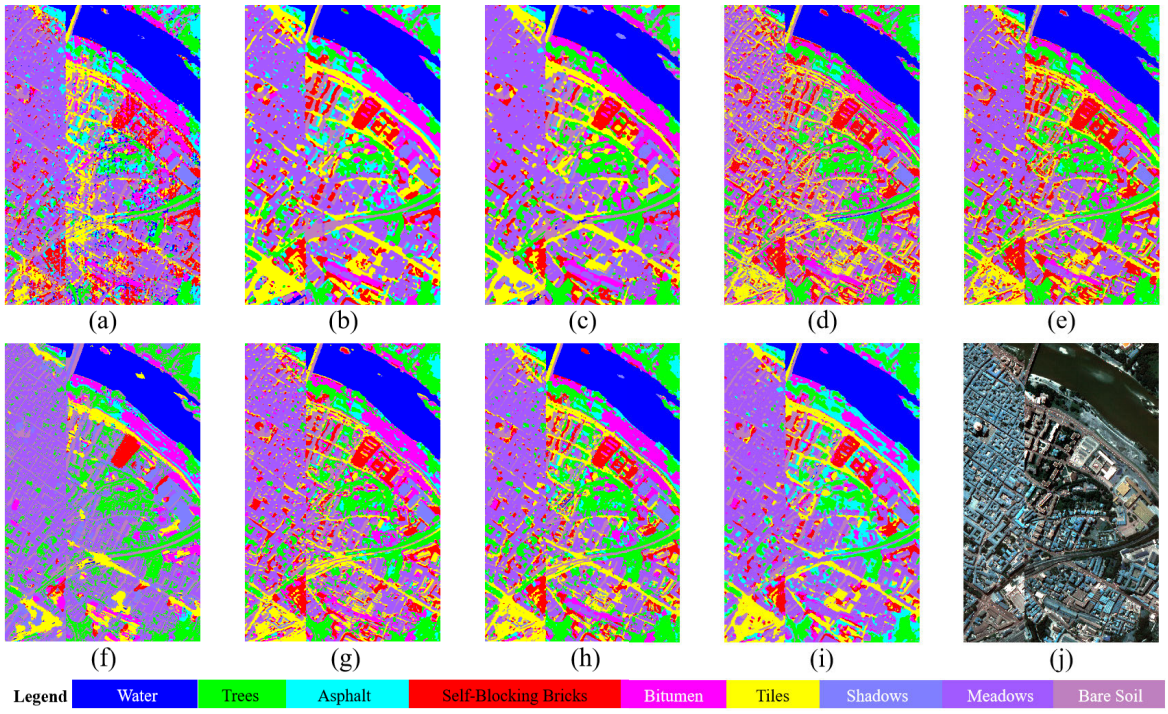


Fig. 11. Full classification maps obtained via various algorithms for the PC dataset. (a) A^2S^2K-RNet. (b) LMSS-NAS. (c) S3Net. (d) DFSL. (e) DCFSL. (f) HFSL. (g) CMFSL. (h) CA-CFSL. (i) HCFSL-NAS. (j) False-color image.

network design of their feature extractors [e.g., Params for the IP dataset are 0.37, 4.26, and 3.30 M, respectively, which are 6, 71, and 55 times higher than HCFSL-NAS (0.06 M)]. In addition, FLOPs are influenced by both the

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TABLE IX
COMPARISON OF COMPUTING RESOURCES OF DIFFERENT ALGORITHMS OF FOUR DATASETS

Data sets	Models	A^2S^2K-RNet [42] 2021-TGRS	LMSS-NAS [37] 2023-TGRS	S3Net [43] 2022-TGRS	DFSL [24] 2019-TGRS	DCFSL [27] 2022-TGRS	HFSL [28] 2022-LGRS	CMFSL [25] 2022-TIP	CA-CFSL [26] 2023-TGRS	HCFSL-NAS
IP	Params/M	0.37	0.11	0.15	**0.03**	4.26	3.30	0.61	0.11	0.06
	FLOPs/G	0.1716	0.0205	0.0765	0.0415	0.0470	0.2006	0.0271	0.0432	**0.0030**
	GPU/M	**1049**	1051	2385	1843	2111	2663	1635	2439	2363
	TST/s	7.54	5.41	8.30	3.31	3.06	8.71	**2.39**	3.91	2.88
	OA/%	64.66	64.92	66.97	55.56	61.31	73.35	65.11	66.34	**77.16**
PU	Params/M	0.22	0.11	0.09	**0.03**	4.26	3.16	0.61	0.10	0.05
	FLOPs/G	0.0879	0.0198	0.0404	0.0531	0.0462	0.1998	0.0271	0.0424	**0.0022**
	GPU/M	**899**	1033	1483	1115	1651	1607	1239	1885	1627
	TST/s	28.09	21.62	27.39	11.57	13.60	41.78	**6.99**	13.46	12.68
	OA/%	63.77	70.92	80.41	74.37	78.72	87.99	82.53	82.36	**88.51**
SA	Params/M	0.38	0.11	0.15	**0.03**	4.26	3.31	0.61	0.11	0.06
	FLOPs/G	0.1751	0.0205	0.0765	0.0415	0.0470	0.2007	0.0271	0.0432	**0.0030**
	GPU/M	**1049**	1053	2399	1845	2123	2585	1493	2397	2353
	TST/s	27.71	27.66	41.79	17.05	16.77	64.24	**12.02**	20.15	16.12
	OA/%	86.82	87.21	90.93	86.89	88.83	89.19	88.52	89.46	**92.40**
PC	Params/M	0.22	0.11	0.09	**0.03**	4.26	3.15	0.61	0.10	0.05
	FLOPs/G	0.0861	0.0205	0.0404	0.0415	0.0462	0.1997	0.0271	0.0424	**0.0022**
	GPU/M	**899**	1033	1501	1115	1569	1547	1191	1775	1613
	TST/s	81.50	76.33	81.44	42.47	39.7	109.71	**26.59**	45.84	41.33
	OA/%	93.90	94.93	96.62	93.02	97.13	96.05	97.22	97.60	**98.01**

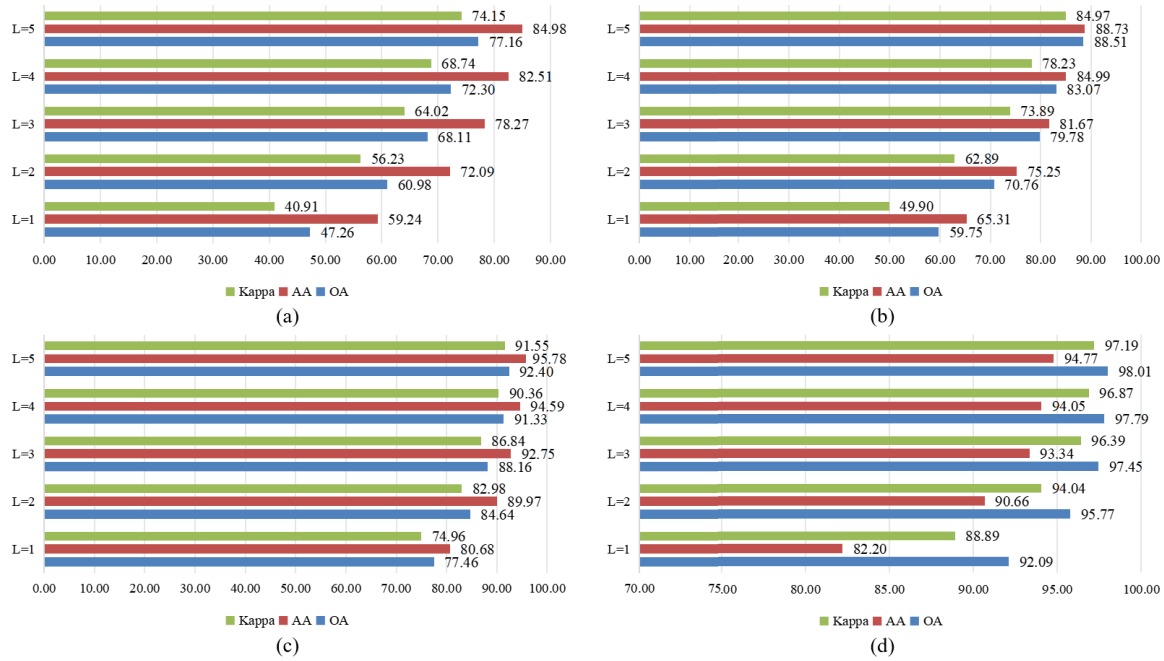


Fig. 12. Classification accuracy of HCFSL-NAS with different numbers of labeled samples on the target-domain dataset. (a) IP. (b) PU. (c) SA. (d) PC.

input data size and Params. Under the same input data size, the FLOPs of A^2S^2K-RNet and DCFSL are at least 15 times higher than that of HCFSL-NAS. Among the FSL-based models, S3Net and DFSL adopt approaches such as spectral band reduction and band selection methods to reduce the computational cost. Specifically, S3Net used a shared-weight S3Net to extract multiscale features and reduce them through principal component analysis. However, the Params remain 1.8–2.5 times higher than that of HCFSL-NAS, and the FLOPs increased at least 18 times due to the multiscale input. Meanwhile, DCFSL, CMFSL, and CA-CFSL combined feature extractors with cross-domain learned features, promoting discriminative feature extraction at the expense of increased computational complexity. In the case of NAS-based methods, the lightweight search space design allows both LMSS-NAS (DL-based) and HCFSL-NAS (FSL-based) to achieve lower Params and FLOPs. However, limited labeled samples affect the performance of LMSS-NAS in supervised learning, leading to overfitting and resulting in poor classification results.

At the same time, the GPU is associated with the batch size, Params, and FLOPs. Therefore, even if Params and FLOPs are low with a small input patch size, a larger batch size will result in a larger CPU load. For example, the FLOPs of HFSL are larger compared to other FSL-based methods under the same batch size. Therefore, it is necessary to consider both model complexity and classification performance for a comprehensive compare and analysis. As observed, in this study, the embedded feature extractor of HCFSL-NAS, which is built based on a lightweight search space design using spatial separable convolutions, requires a lighter number of

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TABLE X
COMPARATIVE EXPERIMENTS (%) OF FUSION LOSS ON FOUR DATASETS

	D_t	IP	PU	SA	PC
HCFSL-NAS_CE	OA	74.25±2.41	84.65±4.36	89.57±1.92	97.50±0.53
	AA	84.14±1.18	86.38±2.91	94.29±1.04	94.26±0.69
	KP	70.96±2.53	80.23±5.03	88.44±2.11	96.46±0.75
	D_t	IP	PU	SA	PC
HCFSL-NAS_SP	OA	52.11±3.00	65.42±4.31	83.94±1.92	89.53±2.03
	AA	66.55±1.87	67.64±4.53	89.20±1.42	81.91±2.23
	KP	46.77±2.95	56.32±4.72	82.18±2.10	85.44±2.70
	D_t	IP	PU	SA	PC
HCFSL-NAS	OA	**77.16±1.39**	**88.51±1.27**	**92.40±0.48**	**98.01±0.37**
	AA	**84.98±1.18**	**88.73±1.65**	**95.78±0.53**	**94.77±0.71**
	KP	**74.15±1.47**	**84.97±1.55**	**91.55±0.54**	**97.19±0.53**

Params, FLOPs, and TST while achieving the best classification performance on four novel datasets.

F. Robustness and Model Regularization

1) Robustness: This study conducted experiments on FSL methods to verify the stability of the models under limited samples. In the experiments, one to five labeled samples were selected per class in the target domain to verify the performance of the models. As shown in Fig. 12, the proposed method showed a steady improvement in classification performance as the number of labeled samples increased from 1 to 5.

2) Model Regularization: To verify the effectiveness of the fusion loss function designed by combining pointwise-based CE loss with pairwise-based SP loss, this study conducted joint ablation experiments. From Table X, it can be observed that HCFSL-NAS significantly improves the performance by learning both the similarities across samples and the accuracy of individual samples. While HCFSL-NAS_CE, only using CE loss as the classification metric function, indicates significant differences in accuracy between different classes and insufficient consistency between the overall sample classification results and the label results, HCFSL-NAS_SP, which uses SP loss as the classification metric function, focuses on the differences between groups and ignores the accuracy of individual samples, resulting in inadequate classification performance in the various target domains.

V. CONCLUSION

This article proposes an automatic FSL algorithm based on NAS for HSIC with few labeled samples. In this research, we first utilize differentiable NAS to automatically search for an optimal feature extractor capable of extracting discriminative features. Second, we leverage the heterogeneous natural image and homogeneous hyperspectral image datasets in the source domain to perform multisource learning. This broader domain learning enhances the effectiveness of metalearning, enabling the algorithm to learn and generalize across different domains. Third, to investigate pixel interactions among the same and different classes, we merge pointwise-based CE loss and pairwise-based SP loss to learn a more robust metric space for accurate classification. Experiments are conducted on four publicly available hyperspectral datasets. The results demonstrate the significant advantages of the proposed method in hyperspectral image few-shot classification tasks. Compared to manually designed methods, our approach achieves higher classification accuracy and exhibits better generalization ability. In conclusion, this article presents a novel approach to address the limited sample problem in HSIC. Future research will focus on exploring the connections between known and unknown categories.

REFERENCES

[1] B. Zhang, "Advancement of hyperspectral image processing and information extraction," *J. Remote Sens.*, vol. 20, no. 5, pp. 1062–1090, 2016.

[2] X. Yang and Y. Yu, "Estimating soil salinity under various moisture conditions: An experimental study," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2525–2533, May 2017.

[3] M. Andrew and S. Ustin, "The role of environmental context in mapping invasive plants with hyperspectral image data," *Remote Sens. Environ.*, vol. 112, no. 12, pp. 4301–4317, Dec. 2008.

[4] C. Zhang, T. Mu, T. Yan, and Z. Chen, "Development and prospect of hyperspectral," *Spacecraft Recovery Remote Sens.*, vol. 39, no. 3, p. 11, 2018.

[5] W. Zhao and S. Du, "Spectral–spatial feature extraction for hyperspectral image classification: A dimension reduction and deep learning approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4544–4554, Aug. 2016.

[6] X. Kang, S. Li, and J. A. Benediktsson, "Spectral–spatial hyperspectral image classification with edge-preserving filtering," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2666–2677, May 2014.

[7] Y. Tarabalka, M. Fauvel, J. Chanussot, and J. A. Benediktsson, "SVM- and MRF-based method for accurate classification of hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 7, no. 4, pp. 736–740, Oct. 2010.

[8] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[9] Y. Chen, X. Zhao, and X. Jia, "Spectral–spatial classification of hyperspectral data based on deep belief network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2381–2392, Jun. 2015.

[10] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, pp. 1–12, Nov. 2015.

[11] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[12] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[13] A. Sellami, M. Farah, I. Riadh Farah, and B. Solaiman, "Hyperspectral imagery classification based on semi-supervised 3-D deep neural network and adaptive band selection," *Exp. Syst. Appl.*, vol. 129, pp. 246–259, Sep. 2019.

[14] J. Li, X. Zhao, Y. Li, Q. Du, B. Xi, and J. Hu, "Classification of hyperspectral imagery using a new fully convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 2, pp. 292–296, Feb. 2018.

[15] L. Windrim, R. Ramakrishnan, A. Melkumyan, and R. J. Murphy, "A physics-based deep learning approach to shadow invariant representations of hyperspectral images," *IEEE Trans. Image Process.*, vol. 27, no. 2, pp. 665–677, Feb. 2018.

[16] S. Jia, S. Jiang, Z. Lin, N. Li, M. Xu, and S. Yu, "A survey: Deep learning for hyperspectral image classification with few labeled samples," *Neurocomputing*, vol. 448, pp. 179–204, Aug. 2021.

[17] S. Mei, J. Ji, Y. Geng, Z. Zhang, X. Li, and Q. Du, "Unsupervised spatial–spectral feature learning by 3D convolutional autoencoder for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6808–6820, Sep. 2019.

[18] Z. Ling, X. Li, W. Zou, and S. Guo, "Semi-supervised learning via convolutional neural network for hyperspectral image classification," in *Proc. 24th Int. Conf. Pattern Recognit. (ICPR)*, Aug. 2018, pp. 1–6.

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[19] Y. Zhan, D. Hu, Y. Wang, and X. Yu, "Semisupervised hyperspectral image classification based on generative adversarial networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 2, pp. 212–216, Feb. 2018.

[20] F. Sung, Y. Yang, L. Zhang, T. Xiang, P. H. Torr, and T. M. Hospedales, "Learning to compare: Relation network for few-shot learning," in *Proc. CVPR*, 2018, pp. 1199–1208.

[21] O. Vinyals, C. Blundell, T. Lillicrap, K. Kavukcuoglu, and D. Wierstra, "Matching networks for one shot learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2016, pp. 3637–3645.

[22] S. Ravi and H. Larochelle. (2016). *Optimization as a Model for Few-Shot Learning*. [Online]. Available: https://api.semanticscholar.org/CorpusID:67413369

[23] J. Snell, K. Swersky, and R. Zemel, "Prototypical networks for few-shot learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 4080–4090.

[24] B. Liu, X. Yu, A. Yu, P. Zhang, G. Wan, and R. Wang, "Deep few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2290–2304, Apr. 2019.

[25] B. Xi, J. Li, Y. Li, R. Song, D. Hong, and J. Chanussot, "Few-shot learning with class-covariance metric for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 5079–5092, 2022.

[26] W. Wang, F. Liu, J. Liu, and L. Xiao, "Cross-domain few-shot hyperspectral image classification with class-wise attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502418.

[27] Z. Li, M. Liu, Y. Chen, Y. Xu, W. Li, and Q. Du, "Deep cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5514505.

[28] Y. Wang et al., "Heterogeneous few-shot learning for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[29] R. Luo, F. Tian, T. Qin, E. Chen, and T.-Y. Liu, "Neural architecture optimization," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, 2018, pp. 7827–7838.

[30] N. Ma, X. Zhang, H.-T. Zheng, and J. Sun, "ShuffleNet V2: Practical guidelines for efficient CNN architecture design," in *Proc. ECCV*, 2018, pp. 116–131.

[31] C. J. Maddison, A. Mnih, and Y. Whye Teh, "The concrete distribution: A continuous relaxation of discrete random variables," 2016, *arXiv:1611.00712*.

[32] J. Mei et al., "AtomNAS: Fine-grained end-to-end neural architecture search," 2019, *arXiv:1912.09640*.

[33] R. Miikkulainen et al., "Evolving deep neural networks," in *Artificial Intelligence in the Age of Neural Networks and Brain Computing*. Amsterdam, The Netherlands: Elsevier, 2019, pp. 293–312.

[34] H. Liu, K. Simonyan, and Y. Yang, "DARTS: Differentiable architecture search," in *Proc. ICLR*, 2018, pp. 1–13.

[35] Y. Chen, K. Zhu, L. Zhu, X. He, P. Ghamisi, and J. A. Benediktsson, "Automatic design of convolutional neural network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 7048–7066, Sep. 2019.

[36] H. Zhang, C. Gong, Y. Bai, Z. Bai, and Y. Li, "3-D-ANAS: 3-D asymmetric neural architecture search for fast hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5508519.

[37] C. Cao, H. Xiang, W. Song, H. Yi, F. Xiao, and X. Gao, "Lightweight multiscale neural architecture search with spectral–spatial attention for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5505315.

[38] Z. Meng, L. Jiao, M. Liang, and F. Zhao, "A lightweight spectral–spatial convolution module for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[39] Z. Lv, X.-M. Dong, J. Peng, and W. Sun, "ESSINet: Efficient spatial–spectral interaction network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5525715.

[40] M. He, B. Li, and H. Chen, "Multi-scale 3D deep convolutional neural network for hyperspectral image classification," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2017, pp. 3904–3908.

[41] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D-2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[42] S. K. Roy, S. Manna, T. Song, and L. Bruzzone, "Attention-based adaptive spectral–spatial kernel ResNet for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7831–7843, Sep. 2021.

[43] Z. Xue, Y. Zhou, and P. Du, "S3Net: Spectral–spatial Siamese network for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5531219.

[44] M. M. Bejani and M. Ghatee, "A systematic review on overfitting control in shallow and deep neural networks," *Artif. Intell. Rev.*, vol. 54, no. 8, pp. 6391–6438, Dec. 2021.

[45] W. Li et al., "Few-shot hyperspectral image classification using meta learning and regularized finetuning," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5529514.

[46] Z. Li, H. Guo, Y. Chen, C. Liu, Q. Du, and Z. Fang, "Few-shot hyperspectral image classification with self-supervised learning," *IEEE Trans. Geosci. Remote. Sens.*, vol. 61, 2023, Art. no. 5517917.

[47] J. Bai et al., "Few-shot hyperspectral image classification based on adaptive subspaces and feature transformation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5523917.

[48] W. Wei, S. Zhao, S. Xu, L. Zhang, and Y. Zhang, "Semi-supervised neural architecture search for hyperspectral imagery classification method with dynamic feature clustering," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5513314.

[49] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2818–2826.

[50] S. Woo, J. Park, J. Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 3–19.

[51] X. Zhou, Y. Zhong, Z. Cheng, F. Liang, and L. Ma, "Adaptive sparse pairwise loss for object re-identification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 19691–19701.

[52] Q. Wang, F. Zhang, and X. Li, "Optimal clustering framework for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5910–5922, Oct. 2018.

[53] K. Sun et al., "A robust and efficient band selection method using graph representation for hyperspectral imagery," *Int. J. Remote Sens.*, vol. 37, no. 20, pp. 4874–4889, Oct. 2016.

[54] Y. Zhong, X. Hu, C. Luo, X. Wang, J. Zhao, and L. Zhang, "WHU-Hi: UAV-borne hyperspectral with high spatial resolution (H^2) benchmark datasets and classifier for precise crop identification based on deep convolutional neural network with CRF," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112012. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0034425720303825

[55] Y. Zhong et al., "Mini-UAV-borne hyperspectral remote sensing: From observation and processing to applications," *IEEE Geosci. Remote Sens. Mag.*, vol. 6, no. 4, pp. 46–62, Dec. 2018.



Fen Xiao (Member, IEEE) received the B.S. and Ph.D. degrees from Xiangtan University, Xiangtan, China, in 2002 and 2008, respectively.

She was a Visiting Scholar at the State Key Laboratory of Pacific Northwest Pacific, Richland, WA, USA. She is currently a Professor with the School of Computer Science and School of Cyberspace Security, Xiangtan University. Her research interests include visual saliency detection, remote sensing image analysis, and Image description.

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Han Xiang received the B.E. degree from Hunan Police College, Changsha, China, in 2020. She is currently pursuing the Ph.D. degree with the School of Computer Science and School of Cyberspace Security, Xiangtan University, Xiangtan, China.

Her research interests include hyperspectral image classification, machine learning, and deep learning.



Chunhong Cao (Member, IEEE) received the B.S. degree in applied mathematics and software from Central South University, Changsha, China, in 1999, and the M.S. and Ph.D. degrees from Xiangtan University, Xiangtan, China, in 2005 and 2017, respectively.

She was a Visiting Scholar with the Department of Computer Science, University of Georgia, Athens, GA, USA. She is currently a Professor with the School of Computer Science and School of Cyberspace Security, Xiangtan University. Her research interests include hyperspectral image processing, machine learning, and compressive sensing.



Xieping Gao (Member, IEEE) received the B.S. and M.S. degrees from Xiangtan University, Xiangtan, China, in 1985 and 1988, respectively, and the Ph.D. degree from Hunan University, Changsha, Hunan, China, in 2003.

He was a Visiting Scholar with the National Key Laboratory of Intelligent Technology and Systems, Tsinghua University, Beijing, China, from 1995 to 1996, and the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore, from 2002 to 2003. He is currently a Professor with Hunan Provincial Key Laboratory of Intelligent Computing and Language Information Processing, Hunan Normal University, Changsha. His research interests include neural networks, evolution computation, and hyperspectral image processing.

Dr. Gao has been a member of the technical committees of several scientific conferences. He is a regular reviewer of several journals.

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